

Enhanced ionic conductivity in borate ester plasticized Polyacrylonitrile electrolytes for lithium battery application



Mehmet Kaynak^a, Abdulmalik Yusuf^b, Hamide Aydın^b, Mehmet Uğur Taşkıran^c,
Ayhan Bozkurt^{a,b,*}

^a Department of Chemistry, Fatih University, Büyükdere-İstanbul 34500 Turkey

^b BioNanoTechnology R&D Center, Fatih University, Büyükdere-İstanbul 34500 Turkey

^c Kaleseramik Çanakkale Kalebodur Seramik San. A. Ş., Çan-Çanakkale 17400, Turkey

ARTICLE INFO

Article history:

Received 4 November 2014

Received in revised form 23 February 2015

Accepted 25 February 2015

Available online 26 February 2015

Keywords:

Poly (acrylonitrile)

Poly (ethylene glycol) borate ester

Polymer electrolytes

Li-ion conductivity

ABSTRACT

The effect of low molecular weight borate ester on the ionic conductivity of polyacrylonitrile (PAN) was systematically investigated by means of measuring conductivity using the impedance technique. Gel polymer electrolyte (SPE) films consisting of PAN as host polymer, LiCF₃SO₃ and LiBF₄ as ionic salts, and poly (ethylene glycol) borate ester (PEGB), ethylene carbonate/propylene carbonate (EC/PC) as plasticizers were prepared by physical blending method. All measurements were performed on electrolyte samples with various contents. Fourier transform infrared spectroscopy (FT-IR) gave evidence of the interactions among PAN, plasticizers and lithium salt. The surface morphology of the electrolytes was investigated by scanning electron microscopy (SEM). Thermal properties of these materials were searched with thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC). Conductivity studies of these blend polymer electrolytes were carried out in the temperature range 20–100 °C. The maximum ionic conductivity values of 1.8×10^{-3} and 1.4×10^{-4} S cm⁻¹ were measured for the film containing 4PAN-10EC/PC-4LiBF₄ and 4PAN-10PEGB-4LiCF₃SO₃, respectively, at room temperature. These films have a good conductivity, better dimensional stability and wide electrochemical stability window. These unique properties make the composite gel membranes suitable for application as electrolyte in rechargeable Li-ion batteries.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

Lithium ion batteries have acquired great importance in the last several decades as the energy storage of portables devices, such as cell phones and laptop computers [1]. However most lithium ion batteries containing liquid electrolytes have potential safety problems, due to the presence of large amount of organic solvents. In order to solve the safety problems, polymer electrolytes have attracted attention [1,2]. Solid polymer electrolytes as well as gel polymer electrolytes have been reported by many groups [3–5]. Solid polymer electrolytes have advantages like high specific energy, high energy density, leak proof, high ionic conductivity, wide electrochemical stability windows, light, solvent free and easy processability [6].

PAN is one of the popular polymers used as gel-type polymer electrolyte for lithium batteries, which generally exhibit about

10^{-3} S/cm of the ionic conductivities [7,8]. PAN-based electrolytes have shown interesting characteristics like high ionic conductivities, thermal stability, good morphology for electrolyte uptake and compatibility with lithium electrodes [9,10]. However its practical application in gel-type polymers is limited by some factors like the poor dimensional stability, the possible hazard due to high amount organic solvent incorporated, and a progressive evaporation of the liquid solvent that may induce a progressive decrease in the ionic conductivity. To minimize these disadvantages, the preparation and properties of the PAN-based solid polymer electrolytes have been reported [7,11,12].

Several methods have been carried out in order to improve the ionic conductivity of the polymer electrolytes as well as their mechanical properties. They consist of synthesizing new polymers, blending two polymers, cross linking two polymers, addition of inorganic inert fillers and addition of salts [6,13]. Another approach to improve the ionic conductivity of a polymer electrolyte is to add small amounts of low molecular weight nonprotic organic solvents such as ethylene carbonate, propylene carbonate, dialkyl carbonate, or their mixtures, which serve as plasticizers. Such solvents

* Corresponding author: Tel.: +90 212 8663300.

E-mail address: bozkurt@fatih.edu.tr (A. Bozkurt).

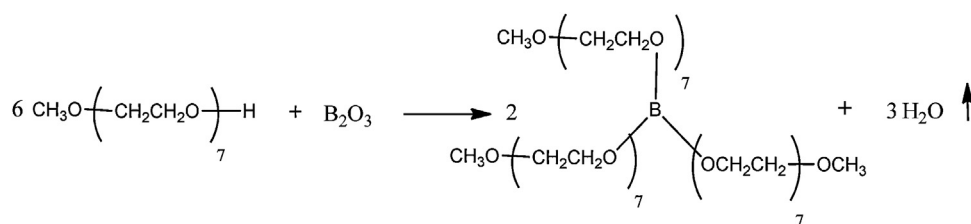


Fig. 1. Synthesis scheme of PEGB.

with high dielectric constants enhance the dissociation and also the solubility of the salt and hence increase the ionic conductivity. However, they impart problems in the applications as they are volatile and flammable [14].

The interest in using of borate esters as additives have increased as they can lead to higher conductivities by trapping the anions due to their Lewis acidic nature, in spite of a very low solvent dielectric constant. They are non-volatile and also have high flash points [15–17]. Many borate esters have been recently synthesized and studied both as additives [18–20] and as solvents [21–25] in salt-in-polymer electrolytes. It is therefore of interest to analyze their influence on conduction of PAN-based electrolytes.

In this work, polymer electrolyte systems of PAN, LiX (X = LiCF₃SO₃ and LiBF₄), EC/PC and PEG borate ester, were prepared. Various additives were used in order to obtain the best compromise between high conductivity, homogeneity and dimensional stability. The polymer electrolytes were characterized via FT-IR, SEM, TGA, DSC and CV. Li-ion conducting properties of the materials were investigated by dielectric-impedance analyzer and the results are discussed and compared with previously reported systems.

2. Experimental

2.1. Materials

PAN (average M_w = 150,000) and Lithium tetrafluoroborate (LiBF₄, 98%) were purchased from Sigma-Aldrich Chemical Company. PEGME (average M_w = 350), Boron oxide (B₂O₃, 99%), Ethylene Carbonate (EC, 99%), Propylene Carbonate (PC, 99%) and Lithium trifluoromethane sulfonate (LiCF₃SO₃, 97%) were purchased from Alfa Aesar Chemical Company. Lithium tetrafluoroborate and Lithium trifluoromethane sulfonate were stored in the glove box.

2.2. Polymer electrolytes synthesis

The PEG borate esters (PEGB) were synthesized, as shown in Fig. 1, by the reaction of methoxy poly (ethylene glycol) (CH₃O (CH₂CH₂O)₇H) (PEGME) and boric acid anhydride (B₂O₃) [26]. The mixture was dissolved in toluene under N₂ atmosphere and refluxed at 110 °C to eliminate the generated water. After the completion of the reaction, toluene was evaporated under vacuum condition, and finally, clear liquid was obtained. The obtained PEGB was preserved in glove box.

Selected amounts of PAN (4 g), plasticizers (10 g PEGB or 10 g EC/PC (1/1) or 5 g PEGME and 5 g EC/PC (1/1)) and LiX (4 g, X = LiCF₃SO₃ and LiBF₄) were taken and dissolved in dimethyl formamide (DMF). Then this solution was stirred continuously for several hours until a homogeneous suspension was obtained. The obtained mixture was cast onto teflon plate. DMF was allowed to evaporate slowly at room temperature and then at 60 °C for the removal of the residual DMF content if any. The obtained film was visually examined for its dryness and free standing nature. The film

was then dried under vacuum at 60 °C for 72 h, followed by storing the film in a glove box.

Solid polymer electrolytes were produced and abbreviated by the amount of used polymer, plasticizers and lithium salt Fig. 2. The thickness of the film was controlled less than 120 μm. All these films can be easily prepared into thin films with good mechanical strength and elasticity.

2.3. Characterizations

Prior to FT-IR spectra measurements, samples were dried under vacuum and stored in a glove box. The IR spectra (4000–400 cm^{−1}, resolution 4 cm^{−1}) were recorded with a Bruker Alpha- P in ATR system.

Thermal stabilities of the polymer electrolytes were examined by thermogravimetry analysis (TGA) with a PerkinElmer Pyris 1. The samples (~8 mg) were heated from room temperature to 750 °C under N₂ atmosphere at a heating rate of 10 °C min^{−1}.

The thermal behavior of the obtained polymer electrolyte samples was investigated by DSC using a Perkin Elmer Pyris 1 instrument. The samples were first cooled from room temperature to −25 °C, kept isothermally for 10 min, and then heated up to 150 °C. After the samples were cooled to −25 °C, and then finally heated to 180 °C. All thermograms were recorded at a rate of 10 °C min^{−1} under nitrogen flow. All DSC experiments were done in duplicate and the thermograms shown refer to the final heating.

The morphologies of NCPE films were observed by scanning electron microscopy (SEM, Philips XL30S-FEG). All of the samples were sputtered with gold for 150 s before SEM measurements.

The alternating current (AC) conductivities of the copolymer samples were measured using a Novocontrol dielectric-impedance analyzer in the frequency range from 0.1 Hz to 3 MHz as a function of temperature. The samples with a diameter of 10 mm and a thickness of approximately 0.4 mm were sandwiched between

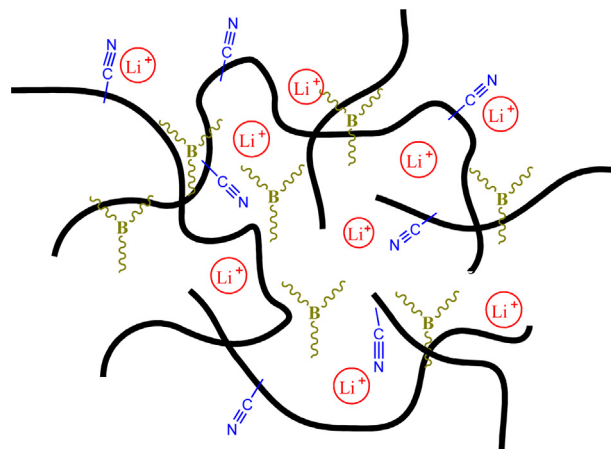


Fig. 2. Illustration of proposed structure of PAN-PEGB based polymer electrolytes.

platinum blocking electrodes and their conductivities were measured at 10 K intervals under a dry nitrogen atmosphere.

Cyclic voltammograms of 4PAN-10PEGME-4LiCF₃SO₃ polymer electrolytes were obtained with a potentiostat Metrohm Autolab (Multichannel-M101 Model). A three-electrode test cell was assembled using stainless steel as the working electrode and Li as the counter and reference electrodes. All the scans were made between 0 and 6 V at a scan rate of 5 mV s⁻¹ at room temperature. Multiple scans were carried out up to 4.4 V and 6 V vs Li/Li⁺ to check stability of the electrolyte.

3. Result and discussion

3.1. Fourier transform infrared spectroscopy

The FTIR spectra of pure PAN and PEGB have been reported in a previous work [27,28]. Characteristic main chain absorptions of the pure PAN are located at 2940, 2243, 1243 and 1072 cm⁻¹, corresponding to C–H, C≡N, C–N and C–C, respectively [31]. PEG borate ester shows strong bands at 2875 cm⁻¹ due to C–H stretching and weak peaks at 667 cm⁻¹ due to B–O units [29]. The absorption bands at ~1029 and ~1350 were assigned to the vibration of B–O–C bond and to the anti-symmetric stretching vibration of B–O, respectively [29].

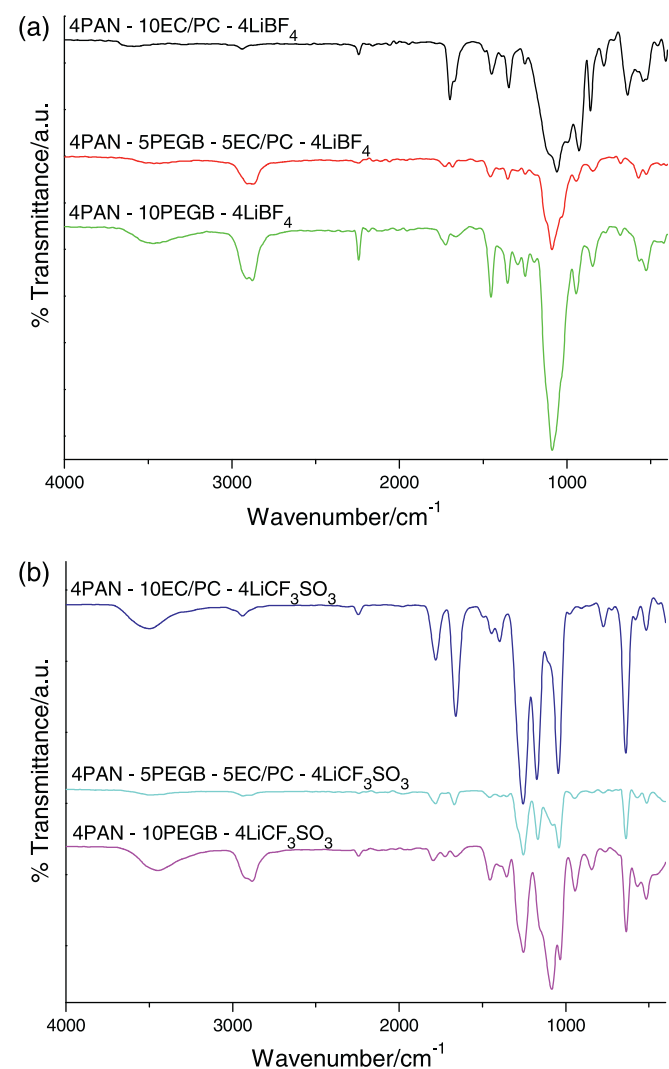


Fig. 3. a Comparisons of FT-IR spectra 4PAN-10PEGB-4LiBF₄ sample. b Comparisons of FT-IR spectra 4PAN-10PEGB-4LiCF₃SO₃ sample.

The FTIR spectra of PAN based polymer electrolytes are shown in Fig. 3a and b. CH stretching frequency of PAN appearing at 2940 cm⁻¹ gets shifted to 2928 cm⁻¹ and PEGB appearing at 2875 cm⁻¹ gets shifted to 2883 cm⁻¹ in the polymer complexes. The band at 2243 cm⁻¹ in PAN is assigned to C≡N stretching vibration. C–N stretching frequency appearing in the pure PAN at 1243 cm⁻¹ is shifted to between 1257 cm⁻¹ in the complex, while C–C stretching vibration at 1072 cm⁻¹ of pure PAN is shifted to 1065 cm⁻¹ in the complex [28,30,31]. Upon addition of the lithium salt, there appeared a higher frequency shoulder in the spectrum. If the N atoms of the C≡N groups are coordinated to Li⁺ ions, the C≡N stretching peak will be shifted to the higher energy side due to an inductive effect [27]. The characteristic frequency at 1790 cm⁻¹ is assigned to the carbonyl stretching of PC in the PC containing electrolytes, the –C=O characteristic frequency of the PC shifts to 1780 cm⁻¹ gradually after the addition of Li salt, indicating the interaction of the plasticizer with Li⁺ on complexation [28].

3.2. Thermal properties

The thermal stability of the PAN based polymer electrolytes with various additives are determined by TGA under N₂ atmosphere from room temperature to 750 °C at a heating rate of 10 °C min⁻¹. The obtained result is shown in Fig. 4.

Physical changes encountered by the polymer electrolyte during thermal excitation has been studied widely using the TG technique, it provides a rough idea of the kind of changes that may occur to the polymer electrolyte when it is applied to working systems. The process involves change in weight attributed to moisture uptake and thermal stability of the polymer electrolyte [31]. Typical thermogram obtained for the films containing PAN based polymer electrolytes are shown in Fig. 4. Clearly, there are three weight loss steps in the thermograms. The first weight loss below 200 °C can be attributed to evaporation of the plasticizers, i.e., EC/PC. The second weight change between 200–350 °C may be due to removal of HCN from acrylonitrile [32]. Final weight loss is due to decomposition of ethylene oxide units of borate esters around 400 °C [15,16]. The decomposition temperature is higher for the sample 4PAN-10PEGB-4LiCF₃SO₃. It was found that the polymer electrolytes containing PEGB whose molecular weight is 350 g/mol, have excellent thermal stability, making them desirable for the development of safe Li-ion secondary battery. Clearly, the polymer electrolytes 4PAN-10EC/PC-4LiCF₃SO₃, 4PAN-10PEGB-

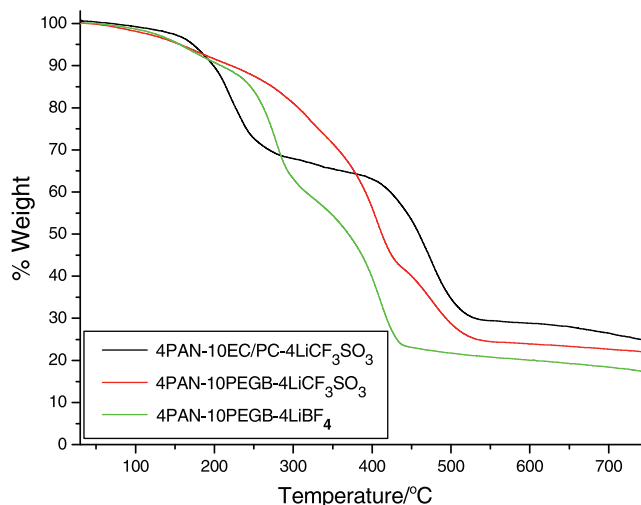


Fig. 4. Thermogravimetry analysis of 4PAN-10PEGB-4LiCF₃SO₃ polymer electrolytes under nitrogen atmosphere at a heating rate of 10 °C min⁻¹.

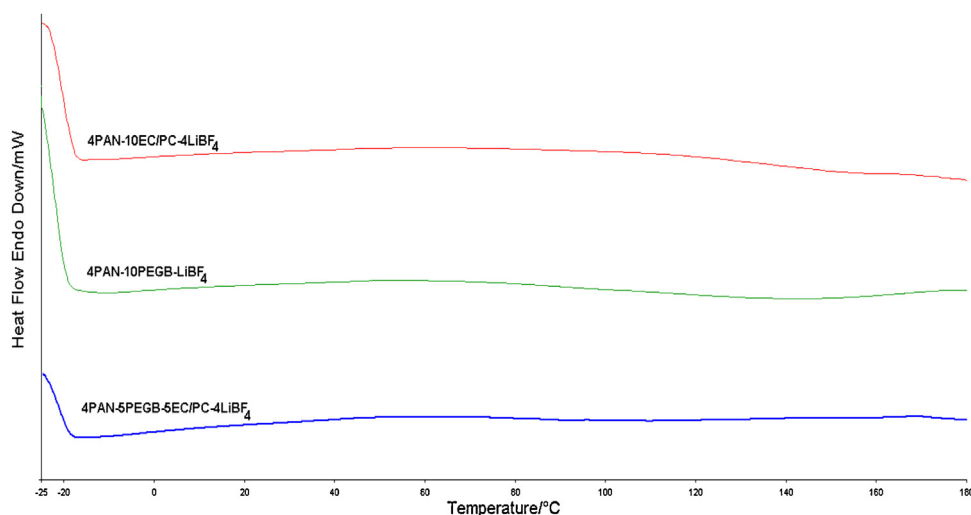


Fig. 5. DSC curves of 4PAN-10PEGB-4LiBF₄ polymer electrolytes. Second heating run endotherms are presented at a heating rate of 10 °C/min.

4LiCF₃SO₃ and 4PAN-10PEGB-4LiBF₄ are found to be stable up to approximately 200 °C.

The thermal properties of the polymer electrolytes were investigated using DSC as well. Fig. 5 shows the melt endotherms of polymer electrolyte samples. For all series, there is no definite glass transition temperature (T_g) within the temperature range of measurements. As the materials are soft at ambient temperature and by embedding the plastifying agents, their glass transition temperatures may be shifted well below −20 °C. Additionally, from DSC results it could be mentioned that 4PAN-10PEGB-4LiBF₄ electrolyte series were fully amorphous.

3.3. Scanning electron microscopy study

The morphology of the samples was characterized by SEM observations. Fig. 6 shows the surface morphology of 4PAN-10PEGB-4LiCF₃SO₃. For this sample, a homogeneous dispersion was observed. It is possible to see a very good dispersion in the films, as there is not a rough area or large agglomeration.

3.4. Ionic conductivity

Lithium ion conductivity of PAN based polymer electrolyte series with different composition of plasticizer and two kind of salt

was measured by the dielectric-impedance analyzer. The films were cut into disks of 10 mm in diameter sandwiched between platinum blocking electrodes and subjected to the impedance analyzer in a completely water-free environment. The ion

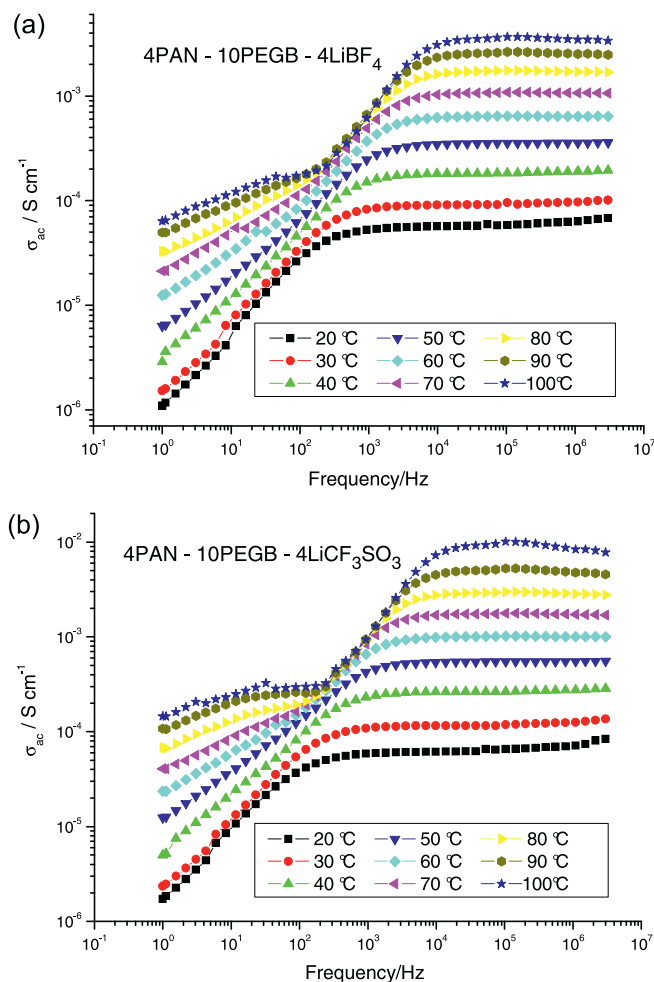


Fig. 7. a AC conductivity (real part) versus frequency of anhydrous 4PAN-10PEGB-4LiBF₄ polymer electrolyte at various temperatures. b AC conductivity (real part) versus frequency of anhydrous 4PAN-10PEGB-4LiCF₃SO₃ polymer electrolyte at various temperatures.

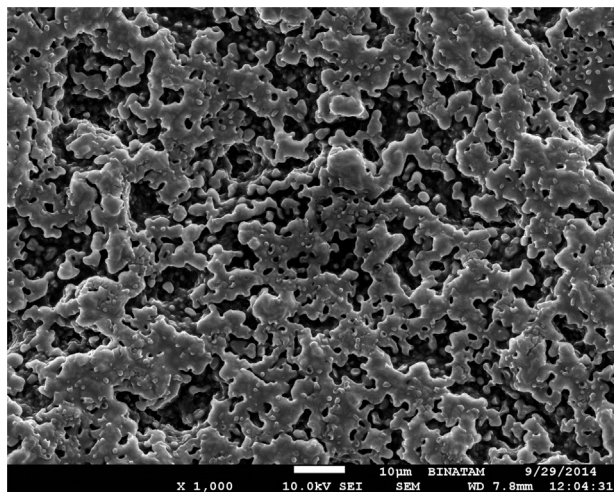


Fig. 6. SEM images of 4PAN-10PEGB-4LiCF₃SO₃ polymer electrolyte sample 10 µm.

dynamics in the polymer electrolytes, the frequency dependence of the conductivity at various temperatures has been analyzed for all samples.

The AC conductivities, $\sigma_{ac}(\omega)$ of PAN based polymer electrolyte membranes with different plasticizers were measured between 20 and 100 °C. The AC conductivities of 4PAN-10PEGB-4LiBF₄ and 4PAN-10PEGB-4LiCF₃SO₃ polymer electrolytes are shown in Fig. 7a and b. Ionic conductivity of all the samples increases with the temperature rise. This well-known phenomenon results from faster ion movement when the temperature increases and as a consequence leads to higher ionic conductivity.

The switchover from the frequency independent to frequency dependent region marks the onset of conductivity relaxation that gradually shifts to higher frequencies as temperature increases. The frequency independent conductivity is, in general, identified with the dc conductivity (σ_{dc}). The DC conductivities of the samples were derived from the plateaus of $\log \sigma_{ac}$ vs. $\log F$ by linear fitting of the frequency independent region of the σ_{ac} . A similar kind of frequency dispersion was observed for all compositions.

The obtained σ_{dc} variations of all samples were compared in Fig. 8 as a function of temperature. The ion conductivity of LiCF₃SO₃ containing polymer electrolytes is higher in the presence of PEGB while the ion conductivity of LiBF₄ containing polymer electrolytes is higher in the presence of EC/PC. The presence of PEGB increases the dissociation of the lithium salt and reduces the aggregation of ions in the polymer electrolytes.

Maximum ionic conductivity was measured for 4PAN-10EC/PC-4LiBF₄ and 4PAN-10PEGB-4LiCF₃SO₃ as $1.8 \times 10^{-3} \text{ S cm}^{-1}$ and $1.4 \times 10^{-4} \text{ S cm}^{-1}$ at 30 °C, respectively. These results are comparable with the literature data [28,30,31]. There are several studies about the ion conductivity property of PAN but most of them are in EC/PC conditions [28,30,31].

The temperature dependence of conductivity values of polymer electrolytes plasticized by borate esters was comparable to the polymer electrolytes plasticized by the mixture of nonprotic organic solvents such as ethylene carbonate, propylene carbonate. The decreased viscosity by the addition of borates seemed to be the one reason for the increase in ionic conductivity. The relatively high conductivity value obtained from the polymer electrolytes plasticized by borate esters, in addition to their superior properties which overcome the drawbacks set by polymer electrolytes containing nonprotic organic solvents makes them better substitutes for lithium battery application.

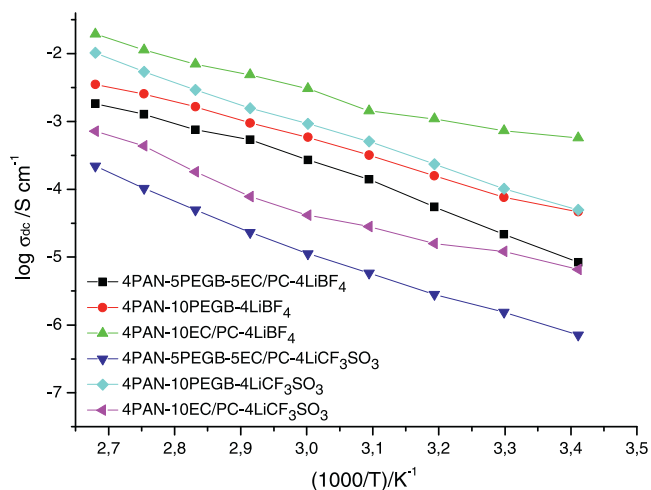


Fig. 8. Temperature dependence of the DC conductivities of PAN based polymer electrolytes with various plasticizers.

3.5. Electrochemical Stability

The electrochemical stability of energy storage devices is one of the important success criteria which offers principal information. The electrochemical stability of the gel electrolytes was evaluated by measuring the cathodic and anodic breakdown potentials. Cyclic voltammetry was performed at room temperature and demonstrated in Fig. 9. The electrochemical stability range of the PAN based electrolytes was determined by running a potential sweep through the system stainless steel/gel electrolyte/Li, in the potential range where the redox reactions for the Li-ion batteries occur.

The performance of the 4PAN-10PEGB-4LiCF₃SO₃ electrolyte was evaluated by multiple potential sweeps up to 4.4 V and 6 V vs Li/Li⁺. It seems that the electrolyte has no the current peak from open circuit potential up to 6.0 V (vs. Li⁺/Li) (Fig. 9a). The current increase above 4.5 V vs Li/Li⁺ may be due to anodic dissolution of stainless steel where the electrochemical stability of the electrolyte is confirmed by limiting the scan to 4.4 V vs. Li/Li⁺ (Fig. 9b). The results showed that the breakdown voltage of the electrolytes, which is regarded as the potential reasons for the decomposition of electrolytes is higher than 6.0 V (vs. Li⁺/Li).

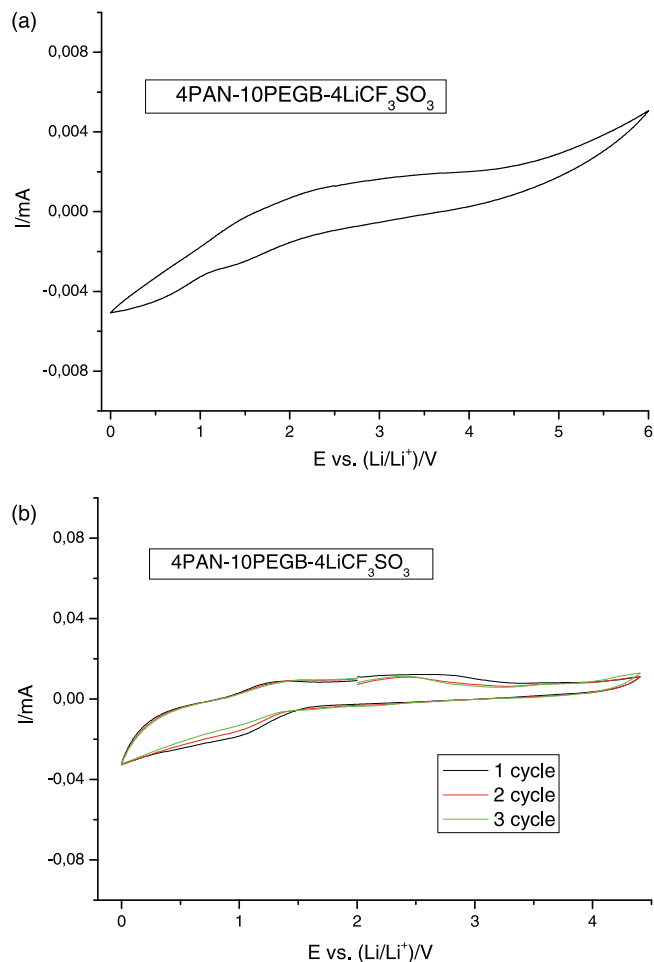


Fig. 9. a Cyclic voltammogram of 4PAN-10PEGB-4LiCF₃SO₃ electrolyte using Lithium reference electrode and counter electrode. Scan rate 5 mV s⁻¹. b Cyclic voltammogram of 4PAN-10PEGB-4LiCF₃SO₃ electrolyte using Lithium reference electrode and counter electrode. Scan rate 10 mV s⁻¹.

4. Conclusions

Blended polymer electrolyte systems consisting of PAN as host polymer, LiCF_3SO_3 and LiBF_4 as ionic salts, PEGB and EC/PC as plasticizers have been prepared by a solvent casting technique. The effect of plasticization on the change in the ionic conductivity and the stability properties has been investigated. The complex formation of the electrolytes has been confirmed from FTIR. Thermal behavior and thermal stability of the electrolytes have also been ascertained with the help of DSC and TGA respectively. The surface morphology of the polymer electrolytes has been studied using SEM. The maximum ionic conductivity of the blended polymer electrolyte 4PAN-10EC/PC-4 LiBF_4 and 4PAN-10PEGB-4 LiCF_3SO_3 is found to be 1.8×10^{-3} and $1.4 \times 10^{-4} \text{ S cm}^{-1}$, respectively, at room temperature. A considerable increase in the ionic conductivity has been observed on plasticization. Hence the properties of the polymer electrolytes seem to be suitable for lithium battery application.

Acknowledgements

This work was supported by The Scientific and Technological Research Council of Turkey (TUBİTAK-TEYDEB-1120047) and partially supported by Turkish Academy of Sciences (TUBA) and National Boron Research Institute (BOREN).

References

- [1] Y.S. Zhu, X.J. Wang, Y.Y. Hou, X.W. Gao, L.L. Liu, Y.P. Wu, M. Shimizu, A new single-ion polymer electrolyte based on polyvinyl alcohol for lithium ion batteries, *Electrochimica Acta* 87 (2013) 113–118.
- [2] Y.P. Wu, X.Y. Yuan, C. Dong, Y.Y. Duan, Lithium Ion Batteries—Applications and Practices, Chemical Industry Press, Beijing, 2011.
- [3] R. Prasanth, V. Aravindan, M. Srinivasan, Novel polymer electrolyte based on cob-web electrospun multi component polymer blend of polyacrylonitrile/poly(methyl methacrylate)/polystyrene for lithium ion batteries—Preparation and electrochemical characterization, *Journal of Power Sources* 202 (2012) 299–307.
- [4] J. Choi, G. Cheruvally, Y. Kim, J. Kim, J. Manuel, P. Raghavan, J. Ahn, K. Kim, H. Ahn, D. Choi, C. Song, Poly(ethylene oxide)-based polymer electrolyte incorporating room-temperature ionic liquid for lithium batteries, *Solid State Ionics* 178 (2007) 1235–1241.
- [5] G. Appetecchi, F. Croce, J. Hassoun, B. Scrosati, M. Salomon, F. Cassel, Hot-pressed, dry, composite, PEO-based electrolyte membranes: I. Ionic conductivity characterization, *Journal of Power Sources* 114 (2003) 105–112.
- [6] S. Ramesh, H. Ng, An investigation on PAN-PVC-LiTFSI based polymer electrolytes system, *Solid State Ionics* 192 (2011) 2–5.
- [7] Y.W. Chen-Yang, Y.T. Chen, H.C. Chen, W.T. Lin, C.H. Tsai, Effect of the addition of hydrophobic clay on the electrochemical property of polyacrylonitrile/LiClO₄ polymer electrolytes for lithium battery, *Polymer* 50 (2009) 2856–2862.
- [8] A.M. Stephan, K.S. Nahm, Review on composite polymer electrolytes for lithium batteries, *Polymer* 47 (2006) 5952–5964.
- [9] A.I. Gopalan, P. Santhosh, K.M. Manesh, J.H. Nho, S.H. Kim, C. Hwang, K. Lee, Development of electrospun PVdF-PAN membrane-based polymer electrolytes for lithium batteries, *Journal of Membrane Science* 325 (2008) 683–690.
- [10] H. Tsutsumi, A. Matsuo, K. Takase, S. Doi, A. Hisanaga, K. Onimura, T. Oishi, Conductivity enhancement of polyacrylonitrile-based electrolytes by addition of cascade nitrile compounds, *Journal of Power Sources* 90 (2000) 33–38.
- [11] H.W. Chen, F.C. Chang, Interaction Mechanism of a Novel Polymer Electrolyte Composed of Poly(acrylonitrile), Lithium Triflate, and Mineral Clay, *Journal of Polymer Science Part B Polymer Physics* 39 (2001) 2407–2419.
- [12] Y.W. Chen-Yang, H.C. Chen, F.J. Lin, C.C. Chen, Polyacrylonitrile electrolytes 1. A novel high-conductivity composite polymer electrolyte based on PAN, LiClO₄ and $\alpha\text{-Al}_2\text{O}_3$, *Solid State Ionics* 150 (2002) 327–335.
- [13] S. Rajendran, R.S. Babu, P. Sivakumar, Investigation PVC/PAN composite polymer electrolytes, *Journal of Membrane Science* 315 (2008) 67–73.
- [14] Y. Karatas, R.D. Banhatti, N. Kaskhedikar, M. Burjanadze, K. Funke, H.-D. Wiemhöfer, Synthesis and Modeling of Polysiloxane-Based Salt-in-Polymer Electrolytes with Various Additives, *Journal of Physical Chemistry B* 113 (2009) 15473–15484.
- [15] Y. Kato, S. Yokoyama, H. Ikuta, Y. Uchimoto, M. Wakihara, Thermally stable polymer electrolyte plasticized with PEG-Borate ester for lithium secondary Battery, *Electrochemistry Communications* 3 (2001) 128–130.
- [16] Y. Kato, K. Hasumi, S. Yokoyama, T. Yabe, H. Ikuta, Y. Uchimoto, M. Wakihara, Polymer electrolyte plasticized with PEG-borate ester having high ionic conductivity and thermal stability, *Solid State Ionics* 150 (2002) 355–361.
- [17] H.S. Lee, X.Q. Yang, C.L. Xiang, J. McBreen, L.S. Choi, The Synthesis of a New Family of Boron-Based Anion Receptors and the Study of Their Effect on Ion Pair Dissociation and Conductivity of Lithium Salts in Nonaqueous Solutions, *Journal of The Electrochemical Society* 145 (8) (1998) 2813–2818.
- [18] X. Sun, H.S. Lee, X.Q. Yang, J. McBreen, A new additive for lithium battery electrolytes based on an alkyl borate compound, *Journal of The Electrochemical Society* 149 (3) (2002) A355–A359.
- [19] S.S. Zhang, C.A. Angell, A Novel Electrolyte Solvent for Rechargeable Lithium and Lithium-Ion Batteries, *Journal of The Electrochemical Society* 143 (12) (1996) 4047–4053.
- [20] W. Xu, X.G. Sun, C.A. Angell, Anion-trapping and polyanion electrolytes based on acid-in-chain borate polymers, *Electrochimica Acta* 48 (2003) 2255–2266.
- [21] S. Tabata, T. Hirakimoto, M. Nishiura, M. Watanabe, Synthesis of a Lewis-acidic boric acid ester monomer and effect of its addition to electrolyte solutions and polymer gel electrolytes on their ion transport properties, *Electrochimica Acta* 48 (2003) 2105–2112.
- [22] M. Saito, H. Ikuta, Y. Uchimoto, M. Wakihara, S. Yokoyama, T. Yabe, M. Yamamoto, Influence of PEG-Borate Ester as a Lewis Acid on Ionic Conductivity of Polymer Electrolyte Containing Mg-Salt, *Journal of The Electrochemical Society* 150 (4) (2003) A477–A483.
- [23] S. Tabata, T. Hirakimoto, H. Tokuda, M.A.B.H. Susan, M. Watanabe, Effects of Novel Boric Acid Esters on Ion Transport Properties of Lithium Salts in Nonaqueous Electrolyte Solutions and Polymer Electrolytes, *Journal of Physical Chemistry B* 108 (2004) 19518–19526.
- [24] Z. Jin, K. Xie, X. Hong, Z. Hu, Capacity fading mechanism in lithium sulfur cells using poly(ethylene glycol)-borate ester as plasticizer for polymer electrolytes, *Journal of Power Sources* 242 (2013) 478–485.
- [25] C.-W. Kuo, C.-W. Huang, B.-K. Chen, W.-B. Li, P.-R. Chen, T.-H. Ho, C.-G. Tseng, T.-Y. Wu, Enhanced Ionic Conductivity in PAN-PEGME-LiClO₄-PC Composite Polymer Electrolytes, *International Journal of Electrochemical Science* 8 (2013) 3834–3850.
- [26] H. Aydın, M. Şenel, H. Erdem, A. Baykal, M. Tülü, A. Ata, A. Bozkurt, Inorganic-organic polymer electrolytes based on poly(vinyl alcohol) and borane/poly(ethylene glycol) monomethyl ether for Li-ion batteries, *Journal of Power Sources* 196 (2011) 1425–1432.
- [27] S. Rajendran, T. Mahalingam, R. Kannan, Experimental investigations on PAN-PEO hybrid polymer electrolytes, *Solid State Ionics* 130 (2000) 143–148.
- [28] X. Helan Flora, M. Ulaganathan, S. Rajendran, Influence of Lithium Salt Concentration on PAN-PMMA Blend Polymer Electrolytes, *International Journal Electrochemical Science* 7 (2012) 7451–7462.
- [29] B. Huang, Z. Wan, G. Li, H. Huang, R. Xue, L. Chen, F. Wang, Lithium ion conduction in polymer electrolytes based on PAN, *Solid State Ionics* 85 (1996) 79–84.
- [30] B.K. Choi, K.H. Shin, Y.W. Kim, Lithium ion conduction in PEO?salt electrolytes gelled with PAN, *Solid State Ionics* 113–115 (1998) 123–127.
- [31] F.A. Amaral, C. Dalmolin, S.C. Canobre, N. Bocchi, R.C. Rocha-Filho, S.R. Biaggio, Electrochemical and physical properties of poly(acrylonitrile)/poly(vinyl acetate)-based gel electrolytes for lithium ion batteries, *Journal of Power Sources* 164 (2007) 379–385.
- [32] A.G. El-Deen, N.A.M. Barakat, K.A. Khalil, H.Y. Kim, Hollow carbon nanofibers as an effective electrode for brackish water desalination using the capacitive deionization process, *New Journal of Chemistry* 38 (2014) 198–205.